

Selenium: A Cost-Effective Test Automation Tool for Web Applications

Selenium is a software testing framework. Test authors can write tests in it without learning a test scripting language. It automates Web based applications efficiently and provides a recording/playback system for authoring tests.



Selenium is a portable software-testing framework for Web applications that can operate across different browsers and operating systems. It is quite similar to HP Quick Test Pro (or QTP, now called UFT) except that Selenium focuses on automating Web based applications. Testing done using this tool is usually referred to as Selenium testing. Selenium is not just a single tool but a set of tools that helps the tester to automate Web based applications more efficiently. It has four components:

1. The Selenium integrated development environment (IDE)
2. The Selenium remote control (RC)
3. WebDriver
4. The Selenium grid

Selenium RC and WebDriver are merged into a single framework to form Selenium 2. Selenium 1 is also referred to as Selenium RC. Jason Huggins created Selenium in 2004. Initially, he named it JavaScriptTestRunner, and later changed this to Selenium. It is licensed under Apache License 2.0. In the following sections, we will learn about how Selenium and its components operate.

The Selenium IDE

The Selenium IDE is the simplest framework in the Selenium

suite and is the easiest one to learn. It is a Firefox plugin that you can install as easily as any other plugin. It allows testers to record their actions as they go through the workflow that they need to test. But it can only be used with the Firefox browser, as other browsers are not supported. The recorded scripts can be converted into various programming languages supported by Selenium, and the scripts can be executed on other browsers as well. However, for the sake of simplicity, the Selenium IDE should only be used as a prototyping tool. If you want to create more advanced test cases, either use Selenium RC or WebDriver.

Selenium RC

Selenium RC or Selenium Remote Control (also known as Selenium 1.0) was the flagship testing framework of the whole Selenium project for a long time. It works in a way that the client libraries can communicate with the Selenium RC server that passes each Selenium command for execution. Then the server passes the Selenium command to the browser using Selenium-Core JavaScript commands. This was the first automated Web testing tool that allowed people to use a programming language they preferred. Selenium RC components include:

1. The Selenium server, which launches and kills the